

NOHIM HASITHA

Computer Science Student | Python, Java & Embedded Systems | AI, Validation Tools & Software Engineering

So(+94)762722301 • nohim.net@gmail.com • <https://www.linkedin.com/in/nohim-hasitha-weedagama-arachchi/>

• [GitHub: github.com/nh7799](https://github.com/nh7799) • Portfolio - <https://nh7799.github.io/portfolioNohim/>

Professional summary

- Computer Science student at the University of Hertfordshire with hands-on experience across Python, object-oriented design, applied AI, and embedded hardware.
- Scored 100% in Java OOP individual coursework and 95% in group coursework, ranking above the cohort mean of 40.21.
- Achieved 84.5% in Artificial Intelligence, implementing grid-world agents and autonomous decision-making algorithms in Python, and integrated this directly to hardware by engineering a custom PCB running an AI agent in MicroPython.
- This combination of software development and embedded systems closely aligns with the challenges I intend to tackle in my future career. Seeking to bring these skills to MediaTek's validation engineering team in Cambridge.
- Experienced in creating dashboards using React.

Skills

Programming: Java • Python • C++ • React • JavaScript • SQL • Linux • Arduino

Embedded systems & Robotics: Raspberry Pi Pico • ESP32/Arduino • sensors • PCB/Schematic design • IoT

Software development: OOP • Debugging • testing • SDLC • Git/GitHub (Version control)

Projects

Embedded validation dashboard (React + Vite)

<https://the-embedded-validation-project.vercel.app/>

01/2026 - 02/2026

- Built and deployed an embedded validation simulator dashboard within a period of 2 days, using React and Vite as development frameworks.
- Designed a user-friendly interface by implementing dashboard summary cards, status filtering, firmware/device metadata, and failure-insight panels designed around embedded software testing workflows.

An artificial intelligence agent based on grid worlds, using a custom-made PCB (agent on PCB)

<https://tinyurl.com/2rjmyj5j>

01/2026 - 05/2026

Designed and built a PCB-based AI agent using MicroPython. Self-taught PCB design via EasyEDA, soldered, assembled the board, and used an OLED for display.

- Gained expertise in PCB design, embedded programming, soldering, and hardware/software integration
- Utilised theoretical understanding of Artificial Intelligence agents into an embedded system within a 1-week period, gained practical knowledge regarding embedded firmware and limitations of an embedded system. Used MicroPython and Thonny as development frameworks to program the microprocessor

Obstacle avoidance robot

<https://www.linkedin.com/in/nohim-hasitha-weedagama-arachchi/details/projects/>

03/2025 - 04/2025

- Built an autonomous robot using Raspberry Pi Pico and MicroPython, implementing sensor-driven movement logic for obstacle detection and avoidance.
- Developed and debugged embedded control software that processed real-time sensor input and converted it into movement decisions.
- Strengthened practical robotics, integration, firmware debugging, and microcontroller development skills through iterative testing.

Education

University of Hertfordshire

2025 - 2026

BSc (Hons) Computer Science

- Relevant modules include Algorithms and Data Structures, Operating Systems and Networks, Artificial Intelligence, Database Systems, and Principles and Practices of Large-Scale Programming.
- Developed practical skills in Java, software design, Linux Operating system, TCP/IP, subnetting, routing, DHCP, databases, debugging, testing, and object-oriented programming.

University of Westminster

2024 - 2025

BSc(Hons) Computer Science

- Completed the first year of Computer Science studies before transferring to the University of Hertfordshire.

Key Achievements

Principles and practices of large-scale programming (Java OOP individual coursework) - (100%)

— Awarded 100% for the coursework with the marker's note citing exceptional user input handling

Artificial intelligence (84.5%) — Accomplished a total module outcome of 84.5% for the artificial intelligence module with a strong foundation in grid worlds and Python.

Principles and practices of large-scale programming (Java OOP Group coursework) - (95%)

— Awarded 95% for group coursework involving object-oriented programming, version control, and collaboration.

Algorithms and data structures (2026) - (88%) — Achieved 88%, strengthening my understanding of problem-solving, algorithmic thinking, and efficient program design.